

The Blackadar–Handelman Choquet-Simplex Conjecture in Stable Rank One

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Status

Literature-implied answer, partial subcase. Conjecture 1.2 in De Silva’s arXiv:1601.03475 is fully settled for every unital C^* -algebra of stable rank one by Antoine–Perera–Robert–Thiel, arXiv:1809.03984, Theorem 4.1. The later authors identify the classical Blackadar–Handelman conjecture, but do not identify De Silva’s paper; the cross-paper implication is therefore recorded as literature-implied.

Source conjecture

On PDF page 1 of *A note on two Conjectures on Dimension functions of C^* -algebras*, Conjecture 1.2 states that the affine space $DF(A)$ of normalized dimension functions is a Choquet simplex for every C^* -algebra A . The source’s Theorem 3.3 verifies this only when the extreme boundary of the quasitrace simplex is finite and either the radius of comparison is finite or $W(A)$ is almost unperforated.

Later theorem

Antoine, Perera, Robert, and Thiel prove in Theorem 4.1 of *C^* -algebras of stable rank one and their Cuntz semigroups* (PDF page 13 of the current arXiv version):

If A is a unital C^* -algebra of stable rank one, then $K_0^*(A)$ is an interpolation group and $DF(A)$ is a Choquet simplex.

The proof first establishes Riesz interpolation for $\text{Cu}(A)$. Stable rank one makes the classical Cuntz semigroup $W(A)$ hereditary in $\text{Cu}(A)$, so interpolation passes to $W(A)$ and then to its Grothendieck group $K_0^*(A)$. The normalized state space of an interpolation group is a Choquet simplex, and this state space is naturally $DF(A)$.

Thus Conjecture 1.2 holds throughout the stable-rank-one class with no simplicity, separability, exactness, finite-radius, or finite-extreme-boundary assumption.

Scope

This theorem does not settle the source’s Conjecture 1.1 concerning density of lower semicontinuous dimension functions, and it does not settle Conjecture 1.2 for arbitrary stable rank. Recent primary-source surveys continue to state the unrestricted Choquet-simplex problem as a conjecture. A separate attempt note records eight routes explored toward a full proof or counterexample and the interpolation/realizability obstructions encountered.

References

- [1] K. De Silva, *A note on two Conjectures on Dimension functions of C^* -algebras*, arXiv:1601.03475.
- [2] R. Antoine, F. Perera, L. Robert, and H. Thiel, *C^* -algebras of stable rank one and their Cuntz semigroups*, Duke Math. J. 171 (2022), 33–99; arXiv:1809.03984.